

Regression Discontinuity Designs

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What's the causal effect of e-voting on the share of valid votes (and downstream, on left-leaning vote share, social spending, etc.)?

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The RD trick: Give up on the ATE; just compare *near the cutoff* where a small variation in municipality population would change its treatment status.

Result

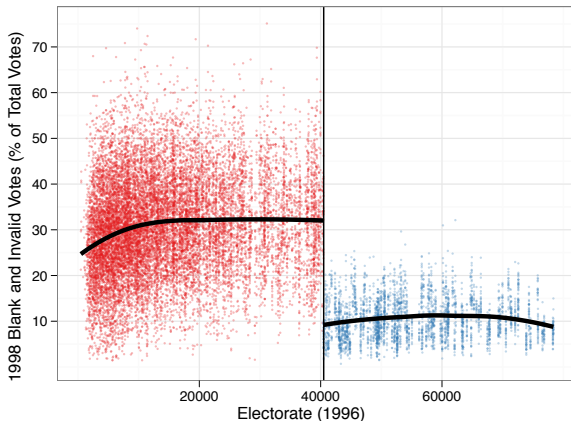


Figure 6: The effect of electronic voting on the percent of null and blank votes. Each dot is a polling station. Polling stations to the left of the vertical black line used paper ballots and polling stations to the right used electronic voting. The black horizontal line is the conditional mean of the outcome estimated with a loess regression.

Generalizing the idea

A binary treatment D_i that is **determined** by whether a continuous predictor X_i crosses a fixed threshold c :

$$D_i = \mathbb{1}\{X_i > c\} \iff D_i = \begin{cases} 1 & \text{if } X_i > c \\ 0 & \text{if } X_i \leq c \end{cases}$$

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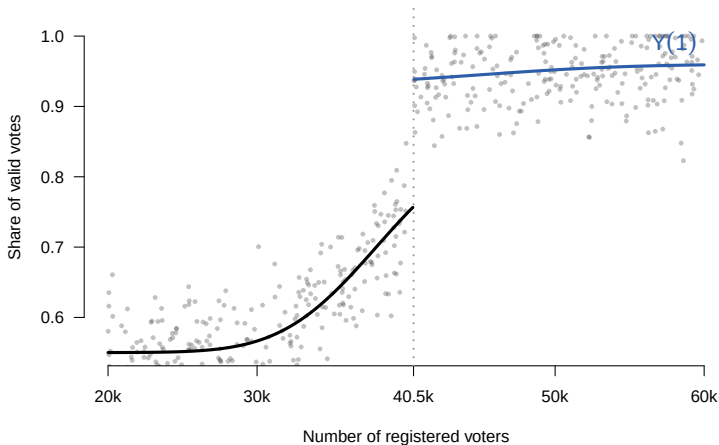
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Later: *fuzzy* RD, where $X_i > c$ **shifts the probability** of D rather than determining it.

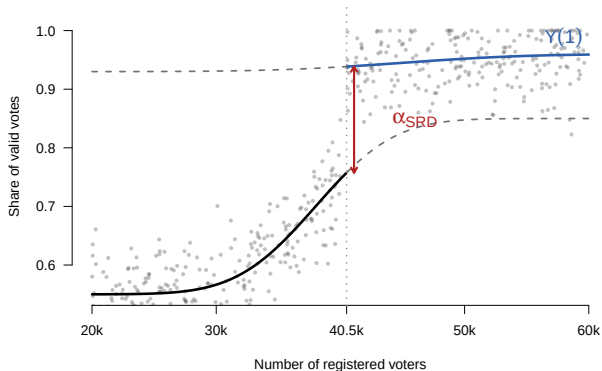
Continuity of potential outcomes

Each municipality has two potential outcomes: $Y(0)$ if paper, $Y(1)$ if e-voting, but we see only one or the other.



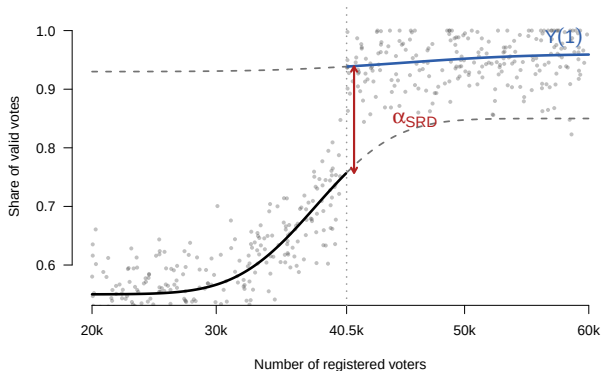
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Really, think of this in terms of two objects: $\mathbb{E}[Y(0) | X]$ and $\mathbb{E}[Y(1) | X]$:



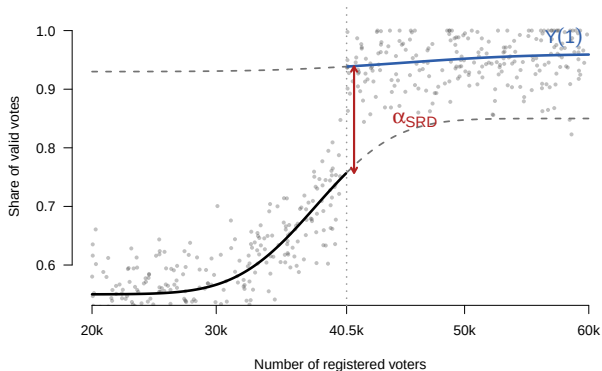
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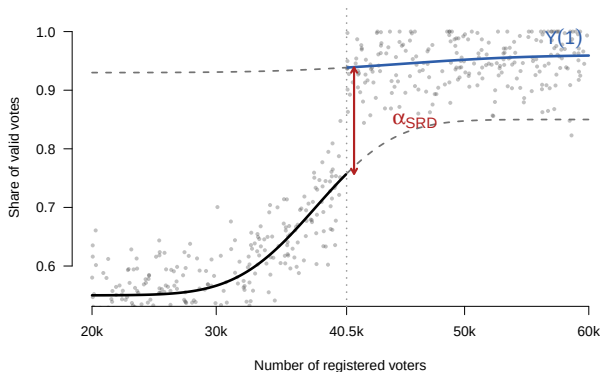
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The effect α_{SRD} is the vertical gap between the tracks *at the cutoff*.

The core assumption is **continuity**: both tracks vary *smoothly* with X at c .

Sharp RD: identification

Identification Assumption (Continuity of conditional expectations)

$\mathbb{E}[Y_{1i} | X_i]$ and $\mathbb{E}[Y_{0i} | X_i]$ are continuous in X_i at $X_i = c$.

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Identification Result

The treatment effect at the threshold is identified as:

$$\begin{aligned}\alpha_{SRD} &= \mathbb{E}[Y_{1i} - Y_{0i} | X_i = c] \\ &= \lim_{x \downarrow c} \mathbb{E}[Y_i | X_i = x] - \lim_{x \uparrow c} \mathbb{E}[Y_i | X_i = x]\end{aligned}$$

*Without further assumptions, α_{SRD} is **only valid at the threshold**.*

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This is *the main* threat to keep in mind. We'll return to it.

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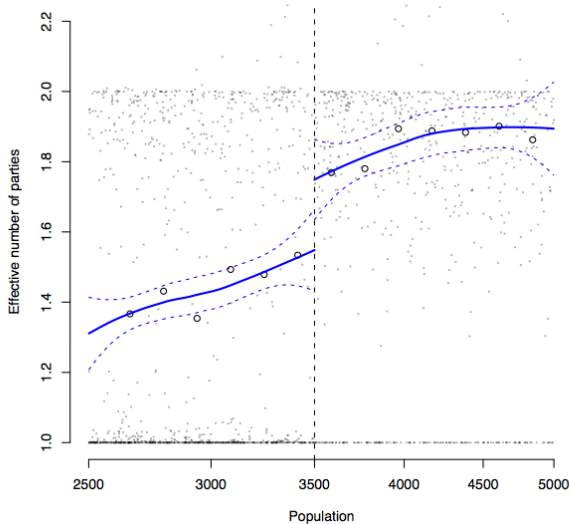
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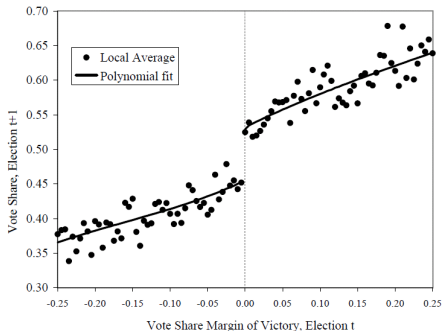
X_i = population, $c = 3,500$, D_i = PR rule, Y_i = number of parties.

Duverger's Law: the jump



Incumbency advantage (Lee, 2008)

Figure IVa: Democrat Party's Vote Share in Election $t+1$, by Margin of Victory in Election t : local averages and parametric fit



- X_i = Dem party vote margin in last election,
- $c = 0$, D_i = won the seat,
- Y_i = Dem party vote share next election.

A few others

- Effect of class size on student achievement (Angrist & Lavy 1999, Maimonides' rule)
- Effect of access to credit (loan offer determined by credit score threshold)
- Effect of mayoral wages on performance (Ferraz & Finan 2011)
- Effect of winning office on politicians' wealth (Eggers & Hainmueller 2009)
- Effect of school district boundaries on home values (Black 1999)
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Warning: "close election" designs with treatment other than *winning* (party, gender, education) measure the effect of *electing somebody with that characteristic* – think districts, not individuals.

How to actually estimate

Historically, or when casually exploring the data, you might try some simple polynomial models:

- Choose a window of the data somewhat near the cutoff to look at.
- Transform your X to be zero at $X = c$, i.e. $\tilde{X} = X - c$.
- Use a linear, quadratic, or cubic model plus a shift at $X = c$; better to allow different slopes on each side
- E.g., $Y \sim \tilde{X} + \tilde{X}^2 + D + \tilde{X}D + \tilde{X}^2D$

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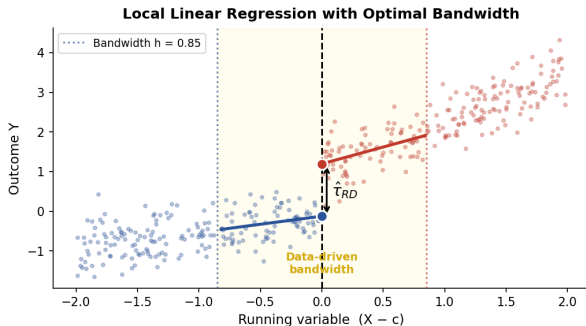
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Both are local: they fit $\mathbb{E}[Y \mid X]$ on each side of c using observations *near* c , and read off the gap at c .

Option 1: `rdrobust` (Calonico, Cattaneo, Titiunik 2014)



- Local linear regression on each side, triangular kernel
- $\hat{\alpha}$ = gap in fitted values at c
- Automated, principled bandwidth selection
- Bias-corrected, robust CIs

`rdrobust(y, x, c = 0)`

Option 2: Gaussian process (`gps` : : `gp_rdd`)

Two risks of any local-polynomial fit:

- Polynomials happily extrapolate near the edge — where they have least data and most leverage.
- Uncertainty should *grow* as we approach c because we have less data around those points.

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The Gaussian process:

- Fits models in a flexible space of smooth functions
- “Carries” an honest range of plausible curve fits through the data → increased uncertainty towards and past edge of the data.
- Regularizes model on each side to avoid misreading a few noisy points as a radically different curve.
- Fully automated (Cho, Kim & Hazlett 2025): `gps::gp_rdd(x, y, cut)`

In practice

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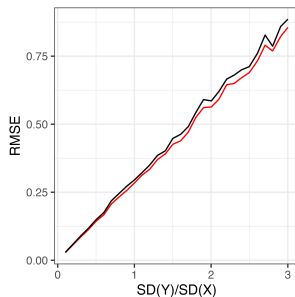
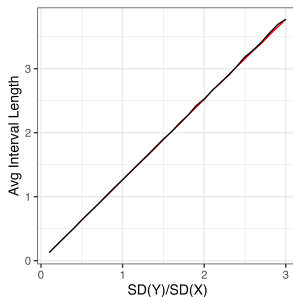
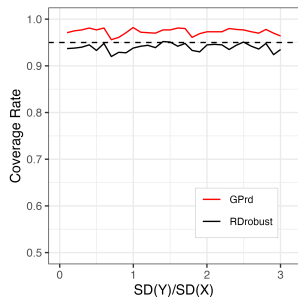
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- Very small samples: `rdrobust` sometimes chooses very wide bandwidth or generates extreme results or NA.

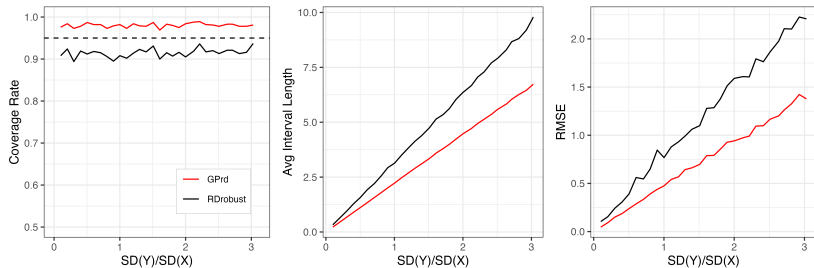
Random Z and Y , large sample

$\tau \in \{0, 3\}$, $N = 500$, `rdrobust` at defaults vs `gp_rdd` at defaults.



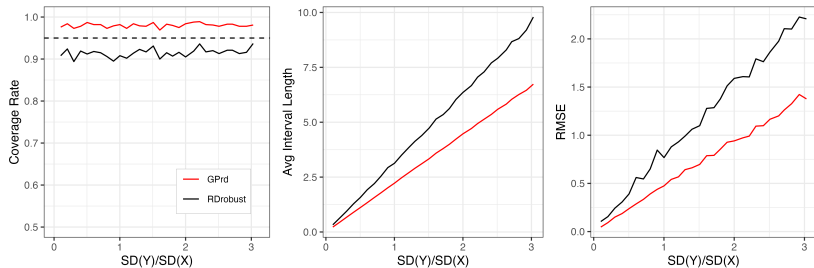
Random Z and Y , small sample

Same DGP, $N = 100$.



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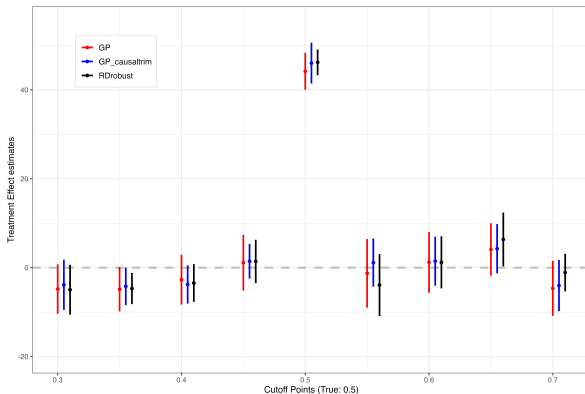
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Improved RMSE is the big story — GP regularizes on each side, where polynomials can go a little crazy.

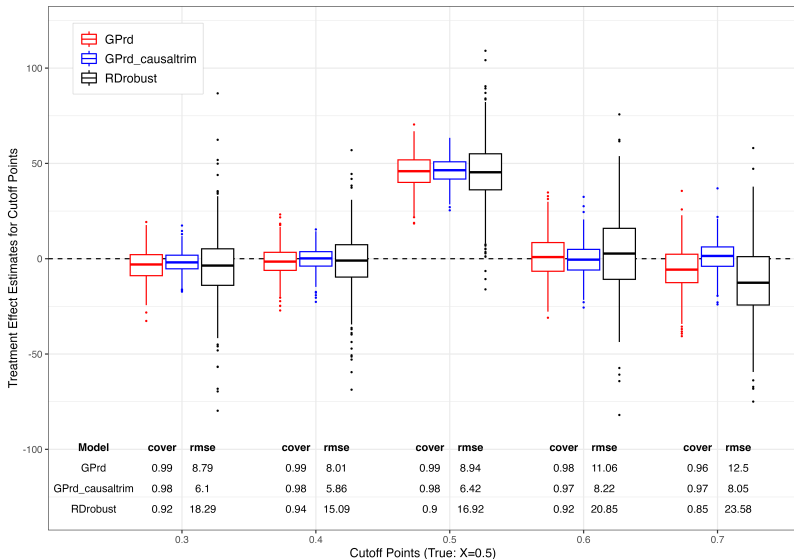
Lee close elections, full sample ($N = 13,577$)

Placebo cutoffs across Lee's close-election data. With this much data, methods barely diverge.

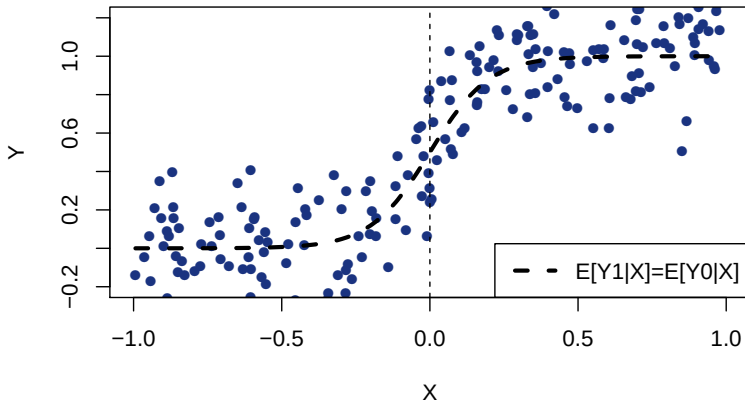


Lee close elections, small subsamples ($N = 200$)

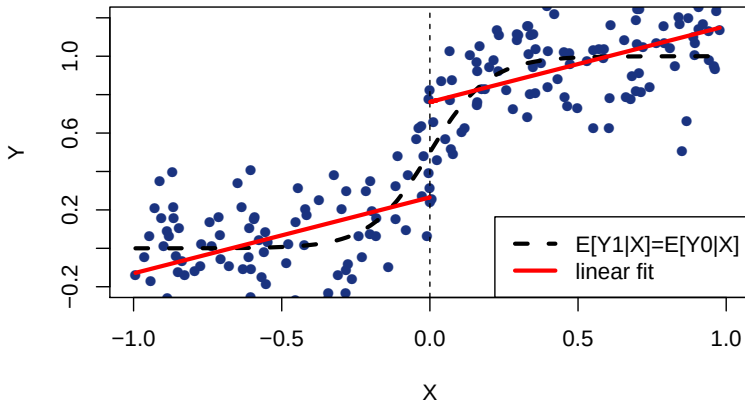
Repeat sampling with $N = 200$ per segment. Now method choice matters.



The risk of misspecification

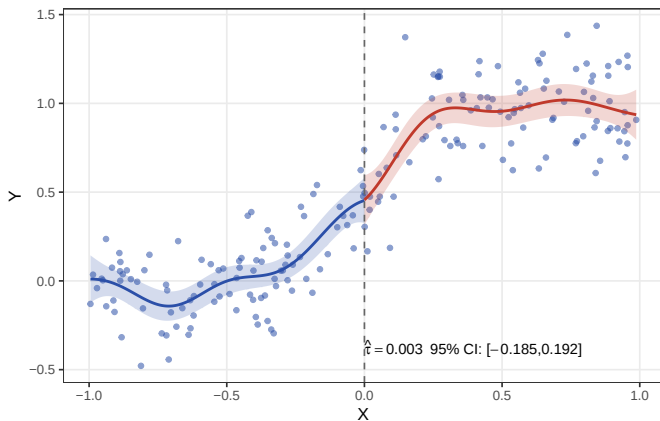


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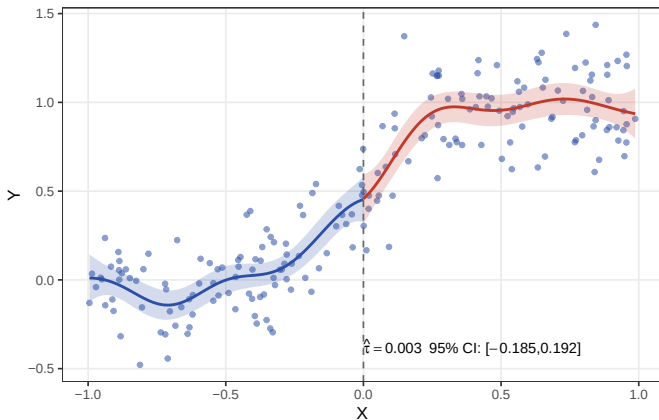
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Should be no problem at all with a suitably flexible estimator.



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gpss::gp_rdd(x, y, cut=0).
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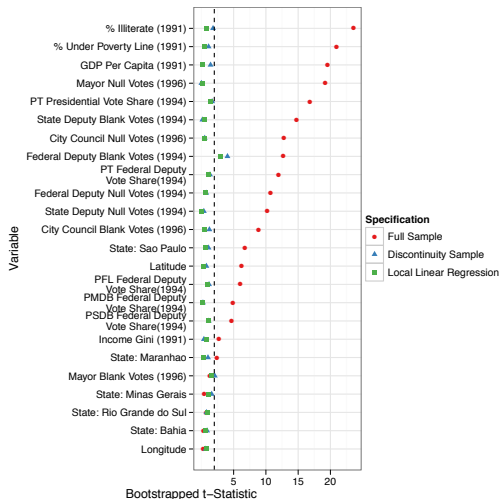
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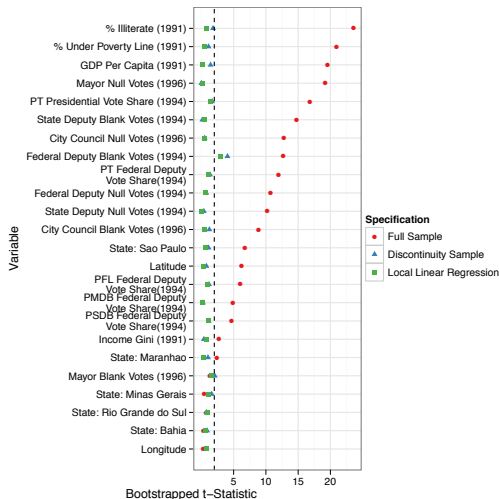
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- ➎ **Sorting:** do units cluster on one side of the cutoff (manipulation)?

Balance test, back to Hidalgo



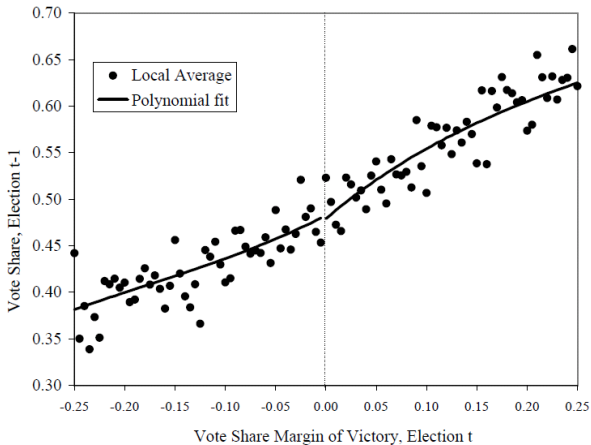
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Run the same RD on a covariate Z as the “outcome” — hope for no jump.

Placebo outcome (Lee 2008)

Most convincing falsification: a covariate (or pre-treatment outcome) plausibly related to potential outcomes but *not* affected by treatment.



Placebo thresholds and sorting

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Sorting: can units *self-select* across the cutoff?

- Fine control over X (gaming a test score, withholding income).
- E.g. Administrators choosing thresholds strategically.
- Density test (McCrary 2008): bulge in X on one side of c ?
- Not always a problem: sorting would have to be very precise, and widespread, to substantially bias the result.

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- In other words, the RD does not tell you *what* treatment effect you are seeing.
- **Always ask:** *what else changes at c ?* This is where careful investigatory work and domain expertise pay off.

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- This should remind you of IV.
 - Crossing c is the instrument.
 - It changes treatment probability and (we hope) only affects Y through treatment.

Doing fuzzy RD

The recipe (just like IV at the cutoff):

- 1 **Reduced form:** estimate the jump in Y at the cutoff (the sharp RD on Y).
- 2 **First stage:** estimate the jump in D at the cutoff (the sharp RD on D).
- 3 Divide.

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`rdrobust` can do this for you (`fuzzy = D`).

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External validity:

- You learn the average effect for units with X_i near c .
- Fuzzy RD: also restricted to compliers near c .
- No information about effects elsewhere on the X distribution. Don't try.
- Often useful in combination with other approaches such as SOO that aim for broader target but have less credible assumptions.

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- Falsification battery: balance/ placebo outcomes, placebo thresholds, sometimes sorting
- But most important is **investigatory work to look for other treatments at same cutoff**.